Jason Vega

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Education

University of Illinois Urbana-Champaign — Urbana, IL
Ph.D. Computer Science — September 2022 - May 2028 (expected)

University of California, San Diego — La Jolla, CA
B.S. Computer Science, GPA: 3.916 (*magna cum laude*) — September 2018 - June 2022

Research Experience

Fine-Tuning Defense Against Multi-Turn Attacks for Large Language Models — Urbana, IL
Graduate Researcher — March 2026 - Present

- **Topic:** Developing a SoTA fine-tuning defense against LLM multi-turn attacks. **Advisors:** UIUC Prof. Gagandeep Singh and collaborators from the Responsible AI team at Amazon.

Rank-Assisted Prefilling Attack and Defense for Large Language Models — Urbana, IL
Graduate Researcher — May 2025 - July 2026

- **Topic:** Exposed how a recent supervised fine-tuning defense to prefilling attacks can be easily circumvented, and proposed a novel attention-based fix. **Advisor:** UIUC Prof. Gagandeep Singh.
- Demonstrated that deep safety alignment (Qi et al., 2025) applied to popular aligned LLMs (Llama, Qwen, Gemma) can be circumvented with just prefilling and access to the top 20 predicted next tokens, and showed that regularizing prefill token attention can mitigate this.
- **Paper: Vega, J.** and Singh, G. (2024). Matching Ranks Over Probability Yields Truly Deep Safety Alignment. In *Trustworthy AI for Good (AI4GOOD) Workshop at ICML 2026*.

Random Augmentation Attacks on Large Language Models — Urbana, IL
Graduate Researcher — March 2024 - January 2025

- **Topic:** Investigated the intersection of safety under random augmentations with multiple dimensions: augmentation type, model size, quantization, fine-tuning-based defenses, and decoding strategies (e.g., sampling temperature). **Advisor:** UIUC Prof. Gagandeep Singh.
- Showed that popular aligned LLMs, such as Llama, Phi and Qwen, can have their safety alignment bypassed in as few as 25 simple random augmentations per harmful prompt.
- **Paper: Vega, J.**, et al. (2024). Stochastic Monkeys at Play: Random Augmentations Cheaply Break LLM Safety Alignment. arXiv preprint arXiv:2411.02785.

Prefilling Attacks on Large Language Models — Urbana, IL
Graduate Researcher — August 2023 - December 2023

- **Topic:** Co-authored the first publication on a simple jailbreaking technique, now known as the *prefilling attack*, to easily bypass LLM alignment. **Advisor:** UIUC Prof. Gagandeep Singh.
- Prefilling is increasingly being used as a baseline attack for evaluating LLM safety; e.g., our paper has been cited in work from Meta (Zhang et al., 2024) and Anthropic (Marks et al., 2025).
- **Paper: Vega, J.***, Chaudhary, I.*, Xu, C.* and Singh, G., Bypassing the Safety Training of Open-Source LLMs with Priming Attacks. In *The Second Tiny Papers Track at ICLR 2024*.

Common Corruptions Robustness of Image Classifiers — Urbana, IL
Graduate Researcher — November 2022 - May 2024

- **Topic:** Investigated training-time methods to improve the robustness of ResNets and Vision Transformers against image corruptions. **Advisor:** UIUC Prof. Gagandeep Singh.
- **Investigated techniques include:** Self-supervised learning (SimCLR) robustness evaluation, regularization of Class Activation Mapping explanations, training a CycleGAN for data augmentation, diversity regularization of ensembles, distillation for robust smaller models

Interpretability Robustness of Image Classifiers — La Jolla, CA
Undergraduate Researcher — January 2021 - June 2022 (Remote)

- **Topic:** Formulating defenses for training robust image classification neural networks against adversarial attacks on various interpretation methods. **Advisor:** UCSD Prof. Tsui-Wei Weng.

- **Contributions:** Implemented defense, verification and attack frameworks, and ran experiments to obtain preliminary robustness results of a 465x improvement compared to standard training.
- **Recognition:** Selected to give a plenary talk to represent the field of Engineering at UCSD's 34th annual Undergraduate Research Conference. Talk available on conference website.

Teaching Experience

UIUC Siebel School of Computing and Data Science Urbana, IL
Teaching Assistant January 2026 - May 2026

- Served as a teaching assistant for CS 444 "Deep Learning for Computer Vision" (∼250 students).
- Engaged with students through virtual office hours and an online class forum, and worked with other TAs to update, test and grade programming homework assignments.

UCSD Computer Science & Engineering Department La Jolla, CA
Course Tutor January 2022 - March 2022

- Tutored in an introductory data structures and object-oriented design course of ∼ 600 students.
- Provided student support through lab interactions and an online classroom forum.
- Managed a pod of 18 students, regularly checking their progress in the course, grading their assignments and intervening when noticing signs of struggle.

Work Experience

Microsoft Bellevue, WA
Software Engineering and Program Management Intern June 2020 - September 2020 (Remote)

- Worked on the new Digital Marketing Center online platform from Microsoft Ads in both program management (weeks 1-6) and software engineering (weeks 7-12) roles.
- Produced a 22 page specification document proposing a new feature, supported by observations from real customer data and with a competitive analysis of four major competitors.
- Implemented a new home page component, including CSS, display logic, responsive layout integration, E2E testing and refractoring of existing code to improve responsive behavior.

Awards

Sloan Scholar University of Illinois Urbana-Champaign
Alfred P. Sloan Foundation's Minority Ph.D. (MPHD) Program (institutional match). Sept. 2022

Wing Kai Cheng Fellowship University of Illinois Urbana-Champaign
A one-year department fellowship graciously sponsored by the Wing Kai Cheng estate. Sept. 2022

Alumni Leadership Scholarship University of California, San Diego
A two-year scholarship awarded for college-level academic and campus leadership. Aug. 2020

Violet and Matthew Lehrer Scholarship University of California, San Diego
A two-year scholarship awarded for college-level academic and campus leadership. Aug. 2020

Academic Services

NeurIPS 2026 ..... Reviewer
ICML 2026 ..... Reviewer
ICLR 2025 ..... Reviewer

Extracurricular Activites

UCSD ACM AI - Event & Social Lead October 2020 - June 2022
Designed and led educational workshops, organized and gave research talks, led research paper reading group sessions and organized social activities for UCSD undergrads interested in artificial intelligence.

Skills

Python, PyTorch, NumPy, Matplotlib, Unix, Conda, LaTeX